

Tech & Teach

Digital Chip Design Program

RTL Design

Task #02

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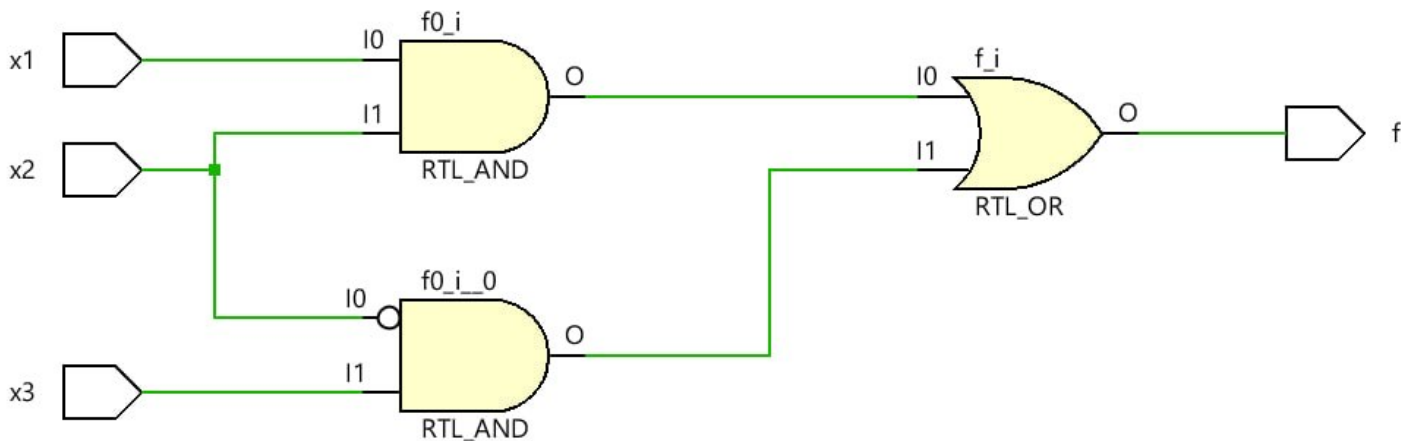
30/6/2026

Data flow

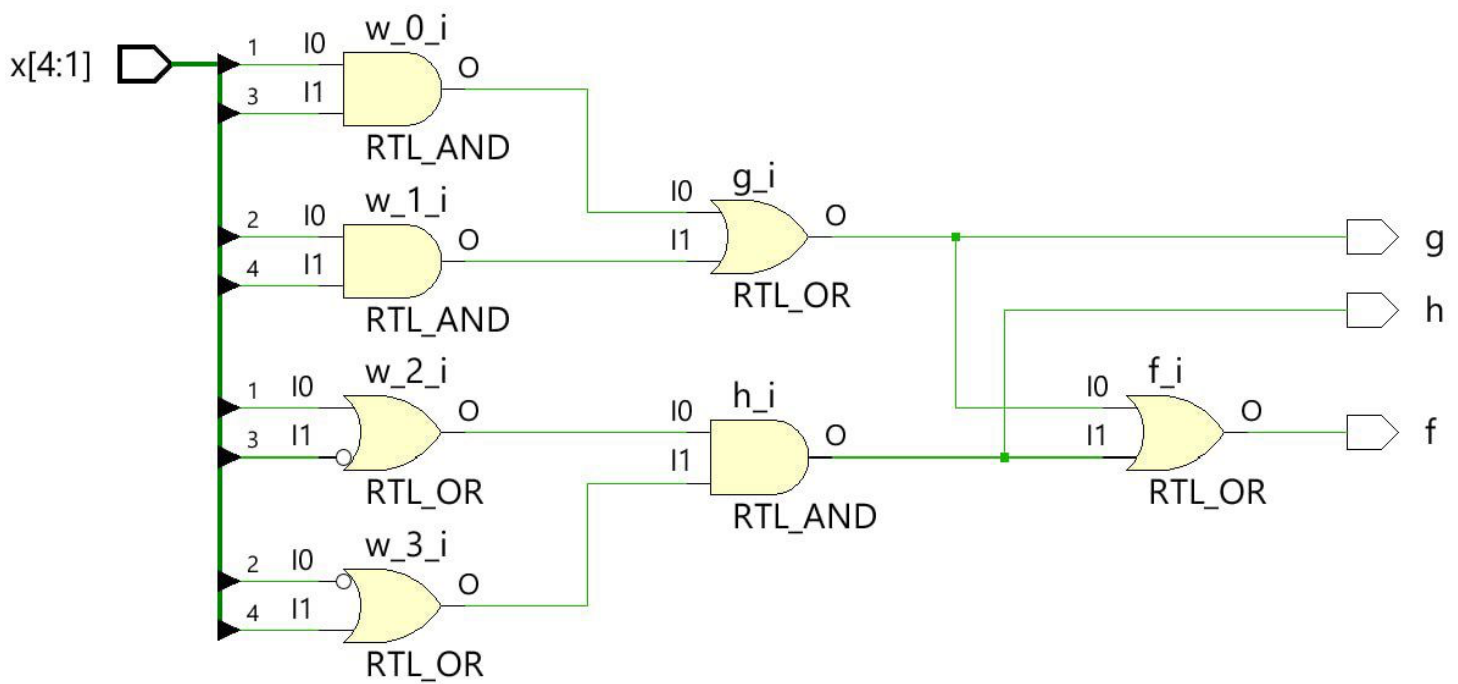
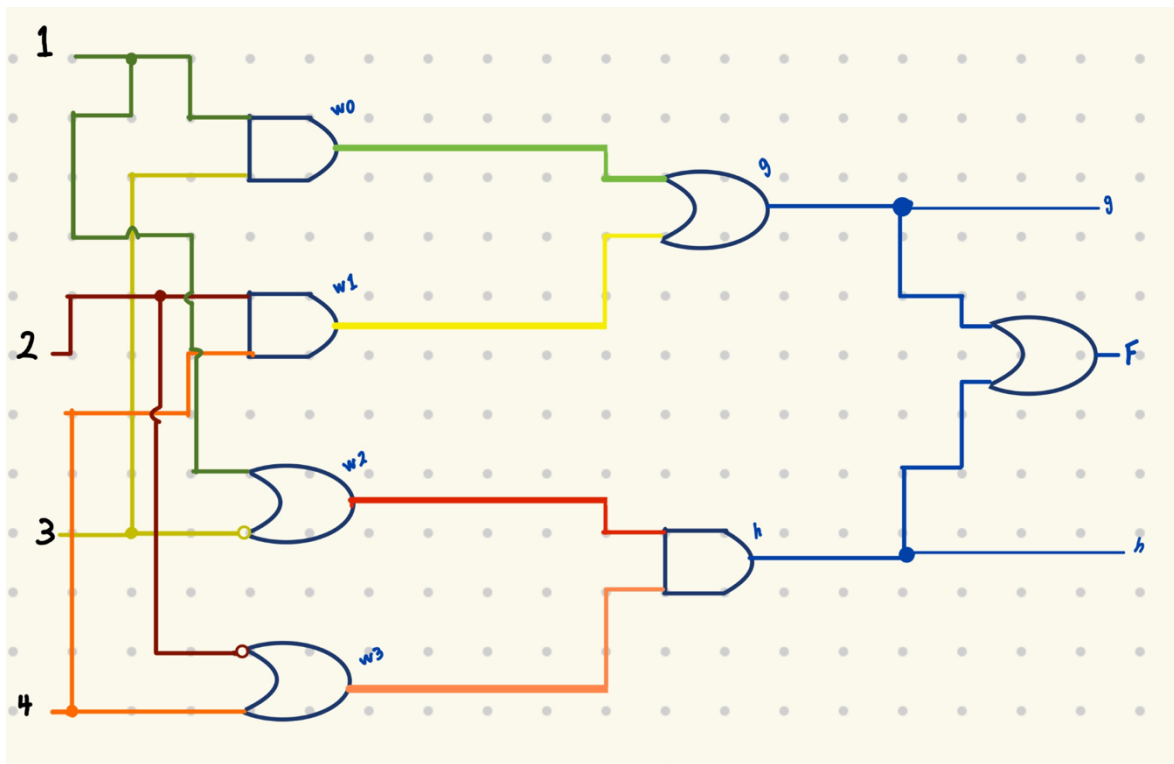
Version 1.0

Task #1.1

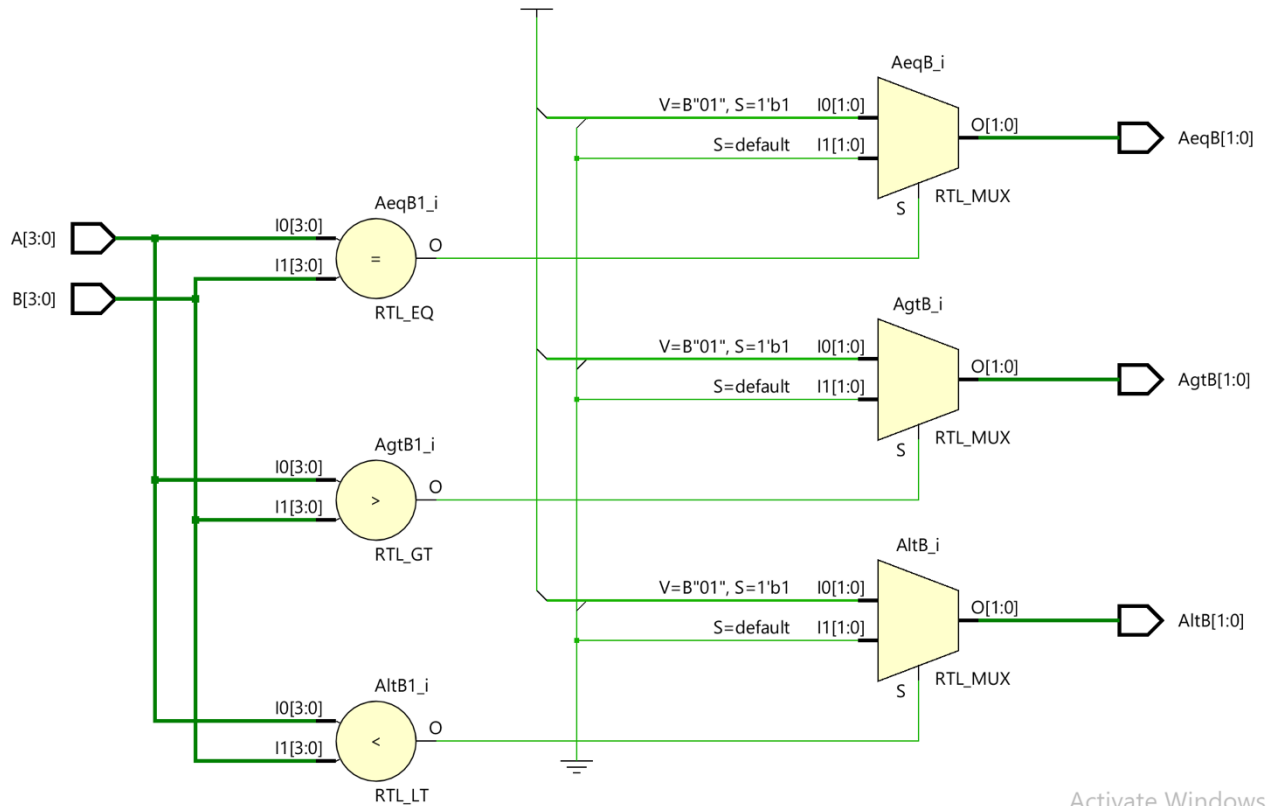
```
2 module lunch(  
3  
4   input logic x1,  
5   input logic x2,  
6   input logic x3,  
7   output logic f  
8   );  
9   assign f = (x1 & x2) | (~x2 & x3);  
10 endmodule
```



Task #1.2



Task #1.3



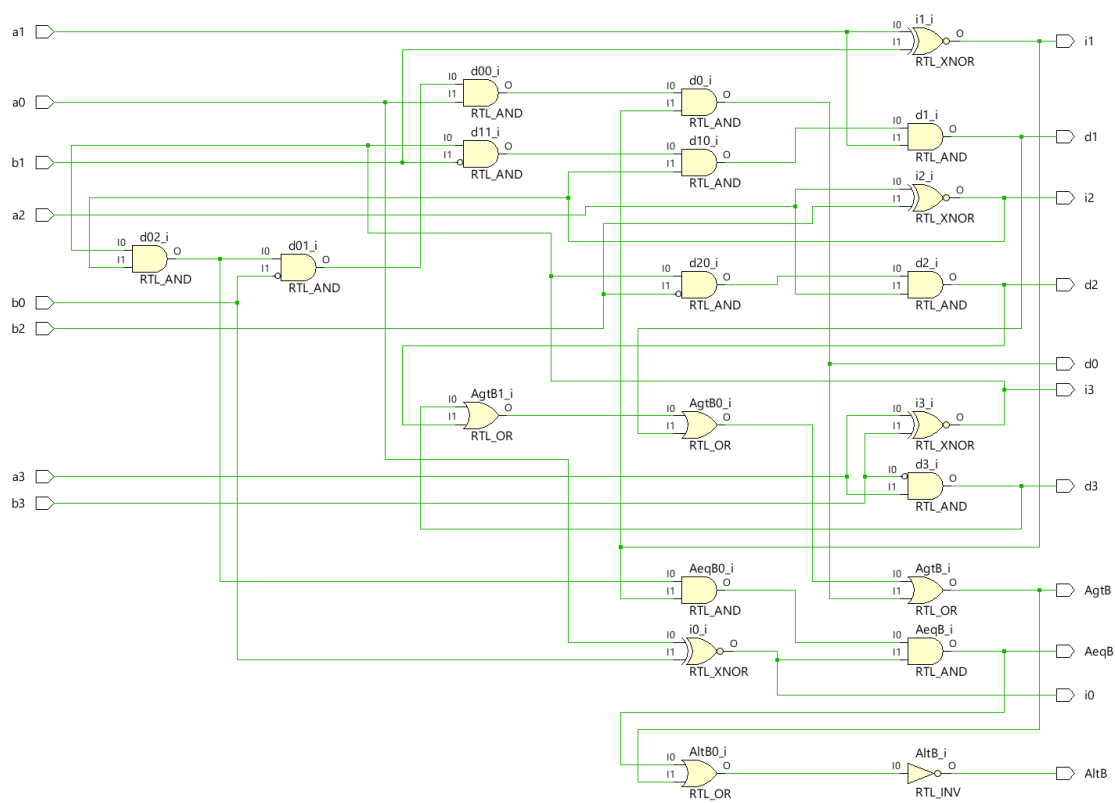
Activate Windows
Go to Settings to activate Windows.

```

1  `timescale 1ns / 1ps
2
3
4  module h (
5      input  logic [3:0] A,
6      input  logic [3:0] B,
7
8          output logic [1:0] AeqB,
9          output logic [1:0] AgtB,
10         output logic [1:0] AltB
11
12 );
13
14 assign AeqB = (A == B) ? 1 : 0;
15 assign AgtB = (A > B)  ? 1 : 0;
16 assign AltB = (A < B)  ? 1 : 0;
17 endmodule

```

ch



```

1  `timescale 1ns / 1ps
2
3  module h(
4      input logic a0,
5      input logic b0,
6      input logic a1,
7      input logic b1,
8      input logic a2,
9      input logic b2,
10     input logic a3,
11     input logic b3,
12
13     output logic i3,i2,i1,i0,
14
15     output logic d3,d2,d1,d0,
16
17     output logic  AeqB,
18     output logic  AltB,
19     output logic  AgtB
20 );
21
22 assign i3 = a3 ^ b3;
23 assign i2 = a2 ^ b2;
24 assign i1 = a1 ^ b1;
25 assign i0 = a0 ^ b0;
26
27 assign d3 = ~b3 & a3;
28 assign d2 = i3 & ~b2 & a2;
29 assign d1 = i3 & ~b1 & i2 & a1;
30 assign d0 = i3 & i2 & ~b0 & a0 & i1;
31
32 assign AeqB = i3 & i2 & i1 & i0;
33 assign AgtB = d3 | d2 | d1 | d0;
34 assign AltB = ~(AeqB | AgtB);
35
36 endmodule

```

ch

Work distribution

Section	Farah&Joud	Tala&Horiah	Rana&Hoor&Sandra
Report	✓	✓	✓
Task 1.1	✓		
Task 1.2		✓	
Task 1.3			✓

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